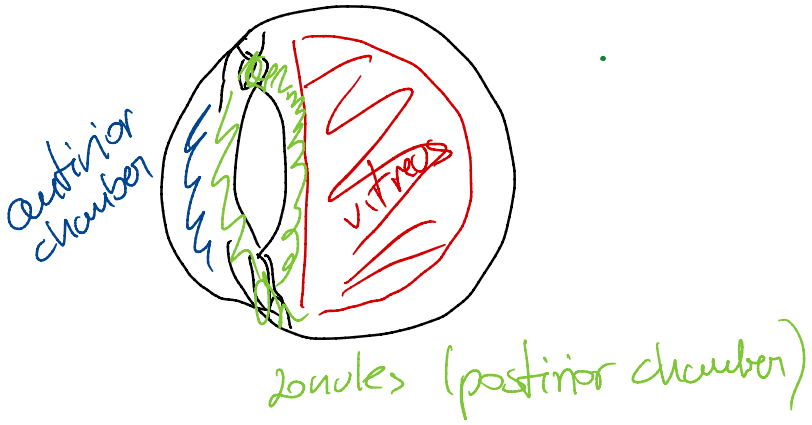
Intraocular Pressure (IOP) and Aqueous Chambers

Images: 6.2, 6.10, 6.3, 6.5, 6.8, 6.9, 6.10

I. Three ocular chambers



A. Vitreous chamber

B. Posterior chamber

C. Anterior chamber - space that hold vitreus

1. Boundaries

a. Anteriorly: corneal endothelium

b. Posteriorly: iris + anterior lens

c. Peripherally: corneoscleral junction (TM, CB, iris root)

2. Anterior chamber angle

a. Aqueous humor exits the anterior chamber.
Where does drains
We have to understand it pretty well, because of glaucoma

b. Scleral spur
dense connective tissue

(1) Dense collagen fibers

(2) Function: Attachment for ciliary muscle fibers

c. Trabecular meshwork
it is  shape

(1) Two anatomic divisions to end Descement membrein (Schwalbes line)

i. Corneoscleral meshwork outer portion

ii. Uveal meshwork inner portion

d. Schlemm canal – black on the pic

(1) Circular vessel that allows aqueous to pass out of the anterior chamber.
even that doesn't have blood in it, but vitreous.

(2) External collector channels empty into deep scleral or intrascleral plexus and then go to the episcleral or conjunctival veins

3. Gonioscopy appearance

a. Gonioscopy uses a special lens that can overcome the total internal reflection. The examiner can then view the structures of the anterior chamber angle.

II. Aqueous dynamics

A. Function of aqueous:

- To provide nutrions to the cornea and lens
- removes/reduces waste products

B. Aqueous production:

by pars plicata (hills and wallys)
in posterior chamber

C. Aqueous outflow

1. Conventional outflow: trabecular flow (conventional way)

through uveal TM → corneal TM → Schlemm's canal → external collector channels → epischlera

2. Unconventional outflow: uveoscleral flow (more recent discovery)

different medicamention for different flow → for GLAUCOMA
goes backwards

uveal TM → ciliary body → superchoroidal space (4 way)

- → sclera
- → ant. ciliary veins
- → suprachoroidal collector channel
- → lymphatic channels